



THE FEDERAL POLYTECHNIC MUBI

BIOLOGICAL SCIENCE TECHNOLOGY

COURSE: Plant Pathology
CODE: STB 315/STB 425
CLASS: HND 1 Biol,
HND 1 Biol./Microbiology &
HND 2 Env. Bio
LECTURER: Ben Toy

LEARNING OBJECTIVES

By the end of this class, the student should be able to:

1. Define the following terms:
 - a. Pathogen
 - b. Pathology
 - c. Plant disease
 - d. Pathogenicity
 - e. Virulence
 - f. Epidemic
 - g. Endemic
 - h. Pandemic
 - i. Sporadic
2. Explain the difference between the following pairs:
 - a. Infectious and Non-Infectious diseases
 - b. Parasites and Saprophytes
 - c. Parasitism and Saprophytism
 - d. Virulence and Pathogenicity
 - e. Pathogen and Pest
 - f. Hyperplasia and Hypertrophy
3. Properly explain the four stages of disease development
 - a. Dispersal
 - b. Penetration (Inoculation)
 - c. Incubation
 - d. Establishment
4. Explain the causes of plant diseases
5. Outline the environmental factors (non-infectious agents) that cause plant diseases
6. Outline the pathogenic (infectious agents) organisms that cause plant diseases
7. Explain the mechanism by which pathogens establish plant diseases
8. Explain the Scope of plant pathology
9. Explain the classification of plant diseases based on:
 - a. Causal agent
 - b. Intensity and Nature
10. Outline and explain the economic importance of plant pathology

INTRODUCTION

Pathogen:

A pathogen is an organism that has the ability to cause or incite a disease condition in a plant.

- (From two Greek words: Pathos = Suffering & Genes = Producing or beginning)
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Pathology:

- (Two Greek words: Pathos = Suffering & Logos = Study)
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Pathology can be simply put as the study of pathogens or the study of diseases

- A plant is said to be healthy or normal when it can carry out its physiological functions to the best of its genetic potential.
 - These physiological functions include;
 - Normal cell division
 - Cell differentiation/development,
 - Absorption of water/minerals from the soil and translocation of these throughout the plant.
 - Photosynthesis and transportation of photosynthetic products to areas of utilization or storage.
 - Metabolism of synthesized compounds.
 - Reproduction and storage of food supplies for overwintering, respiration, etc.
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Whenever pathogens or some other environmental conditions disturb plants and one or more of these functions are interfered with, beyond a certain deviation from the normal, then the plant is said to be diseased.

Plant Pathology

Plant pathology (also phytopathology) is the scientific study of diseases in plants caused by pathogens (infectious organisms) and environmental conditions (physiological factors), their establishment, effects and control.

- Organisms that cause diseases include fungi, oomycetes, bacteria, viruses, viroids, virus-like organisms, phytoplasmas, protozoa, nematodes and parasitic plants.
 - Not included are ectoparasites like insects, mites, vertebrate, or other pests that affect plant health by eating plant tissues.
 - Plant pathology also involves the study of pathogen identification, disease etiology, disease cycles, economic impact, plant disease epidemiology, plant disease resistance, how plant diseases affect humans and animals, pathosystem genetics, and management of plant diseases.
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Plant Disease

Disease in plants then can be defined as any disturbances Brought about by a pathogen or an environmental factor which interferes' with the normal functioning (physiological activities) of the plant in such a way that the affected plant changes in appearance and yields less than a normal healthy plant of the same species and variety.

Pathogenicity,

*Pathogenicity is the quality or state of being **pathogenic**, i.e., the potential ability to produce disease,*

- Pathogenicity is only concerned with “can” or “cannot”.
 - That is the organism can or cannot cause a disease condition on the plant
 - If it can cause a disease condition; the, it is pathogenic.
 - If it cannot cause a disease condition; then, it is non-pathogenic.
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Virulence

*whereas virulence is the disease producing power of an organism, the degree of **pathogenicity** within a group or species.*

- Virulence is concerned with how much of the disease condition, the pathogen can cause
- A pathogen is said to be very virulent if the disease condition it causes is very deadly.

CAUSES OF PLANT DISEASES

- The primary causes of plant diseases are either
 - pathogens or
 - factors in the physical environment.
- The specific mechanisms by which diseases are produced vary considerably, with the
 - Causal agent
 - Plant and
 - Environmental conditions.
- At first, the reaction of the plant to the disease-causing agent at the site of affliction is of a chemical nature and is invisible soon, however, the reaction becomes more widespread and histological changes take place that manifest themselves macroscopically and constitute the symptoms of the disease.
- Affected cells and tissues of diseased plants are usually weakened or destroyed by the disease-causing agent.
- The ability of such cells and tissues to perform their normal physiological functions is reduced or completely eliminated; as a result, plant growth is reduced or the plant dies.
- The kinds of cells and tissues that become infected determine the type of physiological function of the plant that will be interfered with first.
- Thus:
 - infection of the roots interferes with the absorption of water and mineral salts.
 - infection of the foliage interferes with photosynthesis, etc.
 - Infection of the vascular bundles interferes with transport of water, minerals etc.
- In contrast to the above, there is another group of diseases in which the affected cells instead of being weakened or destroyed are stimulated to divide faster (**hyperplasia**) or to enlarge a great deal more than normal cells (**hypertrophy**).

Hyperplasia is excessive cell division causing plant overgrowth resulting in such growths as galls, tumors, enation (an outgrowth from a leaf or stem often associated with viral infection) proliferation, fasciation and witches-broom

Hypertrophy is plant tissue overgrowth caused by abnormal cell enlargement typically caused by pathological enlargement of individual cells in a tissue or organ. For example, some symptoms include puckering and/or curling of leaves.

- Such hyperplastic and hypertrophic cells result in the development of non-functioning abnormal large proliferated organs.
- Overstimulated cells and tissues:
 - Divert much of the available foodstuff to themselves and away from the normal tissues and
 - Interfere with the normal physiological functions of the plant also.
- Diseases caused by environmental factors result from extremes in the conditions supporting life and in the number of chemicals absorbed or required by plants.
- Therefore, for a plant disease to become established, three factors must be in place
 - A pathogen
 - A susceptible host and
 - Conducive environmental conditions
- These three constitute the triangle of plant pathology

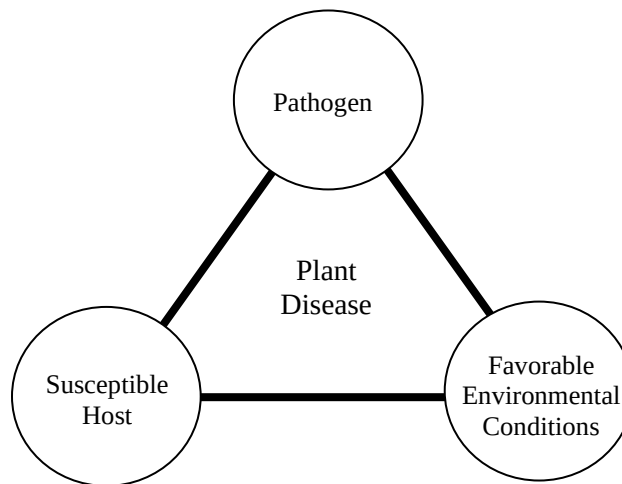


Figure 1: Plant Disease Triangle

SCOPE OF PLANT PATHOLOGY

- Plant pathology can then be looked upon as the study of:
 - The living entities and the environmental conditions that cause diseases to plants.
 - The mechanism by which these factors produce diseases in plants.
 - The interrelationship between the disease-causing agents and the diseased plants.
 - The methods of preventing diseases, alleviating the damage it causes and controlling diseases either after or before they develop in a plant.
- Plant pathologists study the diseases caused by living organisms and disorders caused by excess in balance or lack of certain physical or chemical factors such as moisture, temperature and nutrients.
- Plant damage by insects, man or other animals is not usually included in the scope of study of plant pathology.
- The main purpose of plant pathology is
 - To develop a control for all plant diseases
 - To save the produce which today is destroyed by plant diseases and
 - To make it available to the growers who toil to produce it.

CLASSIFICATION OF PLANT DISEASES

- There are tens of thousands of diseases that affect cultivated plants.
- On average, each kind of crop plant can be affected by one hundred or more plant diseases.
- To facilitate the study of plant diseases, they must be grouped in some orderly fashion.
- This is necessary also for the identification and subsequent control of any given diseases.
- Anyone of several criteria can be used as the basis for the classification of plant diseases.
- Plant diseases are sometimes classified
 - According to the plant organ they affect,
 - According to the symptoms they cause or
 - According to the type of plants affected.

CLASSIFICATION BASED ON CAUSAL AGENT

- However, the most useful criteria for the classification of disease is according to the causative agent of the disease.
- On this basis, plant diseases can be classified into two broad categories.
 - Infectious diseases and
 - Non-infectious diseases

Infectious diseases

These are diseases caused by living organisms or microorganisms also known as Pathogenic Diseases or Pathogenic Infection.

*A **Pathogen** is an organism that can cause or incite a disease in a plant i.e. a **Causal Organism** or an **Incitant**.*

- Some pathogens are parasites while others are saprophytes.

Parasites are organisms that live on or in some other organisms and obtain food from the later but give nothing in return.

- The “Organism” on which the parasite lives on is called the **host**
- The relationship between the parasite and the host is called **parasitism**.

A saprophyte on the other hand is an organism that feeds on dead decaying organic matter.

- Some organisms are obligate **parasites**
 - i.e., they cannot live without the host
 - Most obligate parasites are host specific, i.e., they can parasitize only one species or a variety of plant.
 - Some are even specific to the organs/tissues they attack.
- There are some parasites or pathogens that are not obligate
 - i.e., in the presence of a living host, they become parasites while in the absence of a living host, they are **saprophytes**
 - Such pathogens are called facultative parasites and most of them are not host-specific
 - i.e., they can feed on a wide range of host-plants.
- In many cases, parasitism is intimately associated with pathogenicity.
 - since in most cases, the ability of the parasite to invade and become established in a host generally results in the development of a diseased condition in the host plant.
- The pathogens that cause diseases in plants normally fall into any of the following groups:
 - Fungi
 - Bacteria
 - Mycoplasmas
 - Parasitic plants
 - Viruses and Viroids
 - Nematodes
 - Protozoa

Non-infectious diseases

The non-infectious diseases also known as non-pathogenic diseases or non-parasitic diseases, these are diseases caused by extremes of environmental conditions or unfavourable environmental conditions

- Some of these environmental conditions include:
 - Too low or too high temperatures.
 - Lack or excess of light.
 - Lack or excess of soil moisture.
 - Lack of oxygen.
 - Air pollution
 - Nutrient deficiency.
 - Mineral toxicities.
 - Soil pH
 - Toxicity of pesticides, herbicides, fertilizers, etc.
 - Improper agricultural practices.

CLASSIFICATION OF PLANT DISEASES BASED ON INTENSITY AND NATURE

- Plant diseases can also be classified based on the intensity and nature of disease into the following classes:

- Endemic
- Epidemic
- Sporadic
- Pandemic

Epidemic

- When there is a large case of a particular plant disease happening at the same time,
- Affecting almost every plant in a particular community or a particular environment.

Endemic

- When a plant disease has become established in a community/environment or
- Has become established among a particular species
- Such that its eradication is almost impossible;
- The disease case may be mild or severe.

Pandemic

- A disease condition that spreads over a whole country, region or the whole world.
- Not considering the severity of the disease but the spread.

Sporadic

- This is a disease condition that occurs occasionally or at intervals,
- The intervals may be regular or irregular.
- The disease sometimes may be associated with some seasons and then, seasonal.

DISEASE DEVELOPMENT

- Only infectious diseases will be dealt with in this case.
- This is characterized by the ability of the pathogen to grow, multiply and cause disease;
- Disease development takes place in a number of sequential stages.
 - i. Disease dispersal.
 - ii. Disease penetration.
 - iii. Disease incubation.
 - iv. Disease establishment

DISPERSAL

*The introduction of a pathogen or any part of the pathogen into an area where the pathogen was not there is called **Inoculation***

*Any pathogen or its parts that can cause disease when it is brought into contact with the host is called the **Inoculum**.*

*The movement of the inoculum from an infested/contaminated area to an area free of it is called the **Disease Dispersal**.*

- These inoculum could be whole organisms, fragments of mycelium or spores.
- Dispersal can be done by several agents which may include man, animals, water, farm implements, seeds, etc.

PENETRATION

*Any soil, seeds or plant parts that have the inoculum only in contact with it is said to be **Infested** or **Contaminated**.*

- It is one thing to infest and another to infect
 - i.e. to get into the host.

- There are generally three (3) mechanisms through which pathogens penetrate their host.
 - Through natural openings e.g., stomata and lenticels.
 - Through wounds created by either farm implements, other animals or abrasion with soil.
 - Some organisms penetrate directly by creating the wounds themselves e.g., nematodes.

INCUBATION

This is the period between the penetration of the host by a pathogen and disease establishment i.e., the time during which the pathogen multiplies in number;

- This is mostly affected by:
 - The type of inoculums i.e. whole organisms, fragments of hyphae or spores.
 - The quantity of the inoculums, the higher the quantity, and the shorter the time it takes to get to establish disease.
 - The environmental Conditions

DISEASE ESTABLISHMENT

- This is the stage when the first symptoms of the disease are observed,
- The time when the pathogens overpower the hosts' defense mechanism.
- For a disease to be established, there must be three conditions
 - A susceptible host
 - Conducive environment conditions (for the pathogen)
 - The Pathogen

MECHANISM BY WHICH PATHOGENS ESTABLISH PLANT DISEASE

- Pathogens cause diseases by:
 - Consuming the contents of the host plant.
 - Disturbing the metabolism of the host cells through toxins, enzymes or growth regulatory substances they produce.
 - Weakening the host by continually absorbing food from the host cells for their use and
 - By blocking the transportation of food, mineral nutrients and water through the conductive tissues.
 - Blocking the vascular bundles and hence stopping the transportation of water and other materials in the body of the plant.

ECONOMIC IMPORTANCE OF PLANT DISEASES

- Plant diseases are important to man because they cause damage to plants and plant produce
- For millions of people all over the world who still depend on their plant produce for their existence, plant diseases can make a difference between a happy life and a life haunted by hunger.
- The dead from starvation of the quarter-million Irish people in 1845 and much of the hunger of many people living underdeveloped countries are examples of the consequences of plant diseases.
- Therefore, some of the economic importance of plant diseases are listed thus:
 - i. Plant diseases cause economic losses to growers
 - ii. Plant diseases cause increased prices of produce/products to consumers.
 - iii. Plant diseases destroy the beauty of the environment by damaging plants around homes, along streets, in parks and forests.
 - iv. Plant diseases may limit the kinds of plants that can grow in a large geographical area by destroying all plants of certain species that are extremely susceptible to a particular disease.
 - v. Plant diseases may also determine the kind of agricultural industries and level of employment in an area by affecting the amount and kind of produce available for processing or canning by the industries in the area.
 - vi. On the other hand, plant diseases are responsible for the creation of new industries which develop chemicals, machinery and methods of control of the plant diseases.
 - vii. Financial losses resulting from plant diseases may be incurred indirectly by the farmer having to plant varieties or species of plants that are resistant to the disease but less productive or costlier or commercially less profitable.
 - viii. By having to spray or control a disease, the farmer incurs more expenses.
 - ix. Plant diseases thus can affect the general economy of a country.

ASSIGNMENT

Write an essay on the role of molecular biology in the control of plant diseases

SUBMISSION

On or before 31st July 2023